

Line diagram standard

Issue 4

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Foreword

Line diagrams have one of the most important roles within the Transport for London (TfL) information system. They direct the travelling public around our networks from departure point to destination and confirm their chosen route.

Information of this kind must be easy to understand and clear to read. This document sets out the standards to be followed when producing line diagrams on behalf of TfL.

More information on TfL design rules are available via the [TfL website](#).

I Typography

Our typeface is Johnston, which should be set in mixed upper and lower case.

Johnston 100 Medium is the only typeface used on all line diagrams within the passenger environment, as well as signing and a wide range of publicity and other material.

ABCDEFGHIJKLMNOPQRSTUVWXYZ

abcdefghijklmnopqrstuvwxyz

1234567890£/.,“()::

Johnston 100 Medium

2 Mode colours

The colours shown here are those used to identify each mode.

The Pantone Matching System is to be used for print purposes and the Natural Colour System (NCS) is to be used for paint applications.



Transport for London
PMS 072
CI00 M97 Y3 K3
R0 G25 B168
NCS S 4060-R80B



DLR
PMS 326
C86 M2 Y41 K0
R0 G175 B173
NCS S 2050-B50G



Elizabeth line
PMS 266
C67 M83 Y0 K0
R96 G57 B158
NCS N/A



London Buses
PMS 485
C6 M98 Y100 K1
R220 G36 B31
NCS S 1085-Y80R



**London Cable Car
(IFS Cloud Cable Car)**
PMS N/A
C65 M81 Y0 K0
R115 G79 B160
NCS N/A



London Coaches
PMS 130
C2 M38 Y100 K0
R255 G166 B0
NCS S 1070-Y20R



London Dial-a-Ride
PMS Pantone Purple
C35 M88 Y0 K0
R192 G40 B185
NCS S 2060-R40B



London Overground
PMS 158
C3 M66 Y99 K0
R250 G123 B5
NCS S 0585-Y50R



London River Services
PMS 299
C81 M18 Y0 K0
R3 G155 B229
NCS S 2060-B



London Trams
PMS 368
C59 M2 Y100 K0
R95 G181 B38
NCS S 0580-G30Y



London Underground
PMS 072
CI00 M97 Y3 K3
R0 G25 B168
NCS S 4060-R80B



Santander Cycles
PMS N/A
C0 M93 Y100 K0
R239 G56 B36
NCS N/A

3 London Underground line colours

This section provides the colour references for all the London Underground lines.



Bakerloo line
PMS 470
C26 M70 Y97 K16
R178 G99 B0
NCS S 4050-Y50R



Central line
PMS 485
C6 M98 Y100 K1
R220 G36 B31
NCS S 1085-Y80R



Circle line
PMS 116
C0 M18 Y100 K0
R255 G200 B10
NCS S 0580-Y10R



District line
PMS 356
C96 M27 Y100 K15
R0 G125 B50
NCS S 3065-G10Y



**Hammersmith
& City line**
PMS 197
C3 M48 Y15 K0
R245 G137 B166
NCS S 0550-R10B



Jubilee line
PMS 430
C55 M41 Y38 K5
R131 G141 B147
NCS S 4005-R80B



Metropolitan line
PMS 235
C41 M100 Y41 K21
R155 G0 B88
NCS S 4050-R30B



Northern line
PMS Black
C0 M0 Y0 K100
R0 G0 B0
NCS S 9000-N



Piccadilly line
PMS 072
C100 M97 Y3 K3
R0 G25 B168
NCS S 4060-R80B



Victoria line
PMS 299
C81 M18 Y0 K0
R3 G155 B229
NCS S 2060-B



Waterloo & City line
PMS 338
C55 M0 Y39 K0
R118 G208 B189
NCS S 1040-B80G

4 London Overground line colours

This section provides the colour references for all the London Overground lines.



Liberty line
PMS 6215
C50 M40 Y40 K40
R93 G96 B97
NCS S 6502-B



Lioness line
PMS 2012
C0 M40 Y100 K0
R250 G166 B26
NCS S 1080-Y20R



Mildmay line
PMS 2383
C85 M40 Y5 K10
R0 G119 B173
NCS S 3050-R90B



Suffragette line
PMS 6171
C65 M0 Y75 K0
R91 G189 B114
NCS S 2050-G20Y



Weaver line
PMS 689
C30 M80 Y20 K35
R130 G58 B98
NCS S 4040-R30B



Windrush line
PMS 1795
C0 M100 Y90 K0
R237 G27 B0
NCS S 1080-Y90R

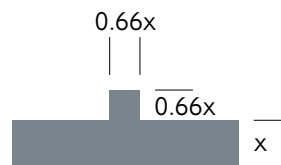
5 Station names and line thickness

The construction of a TfL line diagram is based around the 'x' height of a station or stop name.

This 'x' height is equal to the thickness of the route line. The principles of line spacing are also based upon the 'x' height value as shown.

Station ticks are 0.66x squared.

On the London Underground Tube map, non-Underground lines which would normally be shown as solid lines (such as the DLR, Elizabeth line and London Overground lines) are to be displayed as parallel lines.

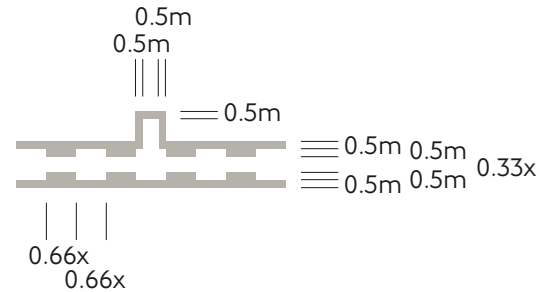


5.1 Station names and line thickness continued

When a section of line is to be displayed as being out of use due to engineering works, the example shown here is to be followed.

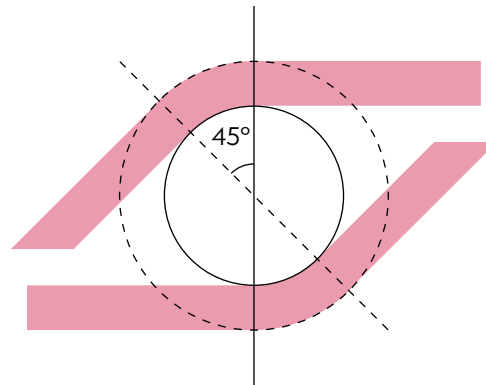
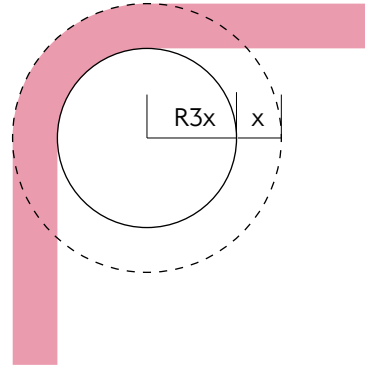
'm' is equal to $0.33x$.

The line colour is always to be displayed in black 30 per cent.



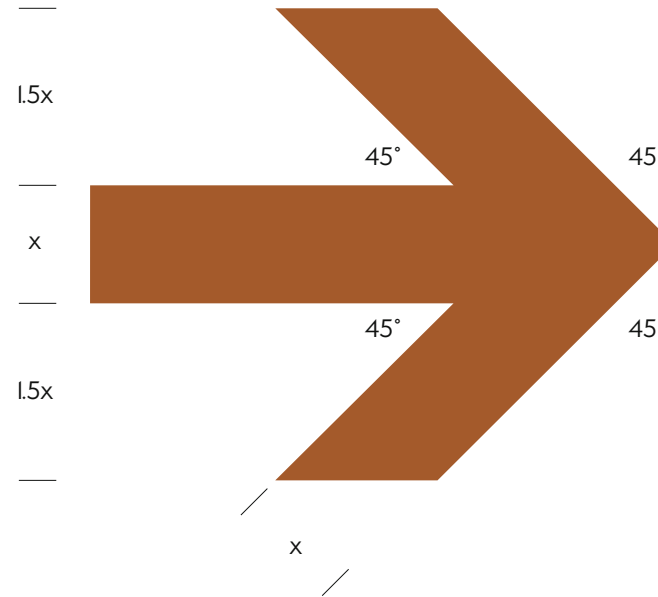
6 Radii and angles

Where a change of direction is necessary, the line must be drawn at an angle of 45 or 90 degrees. Where lines run adjacent, $3x$ is applied to the innermost curve.



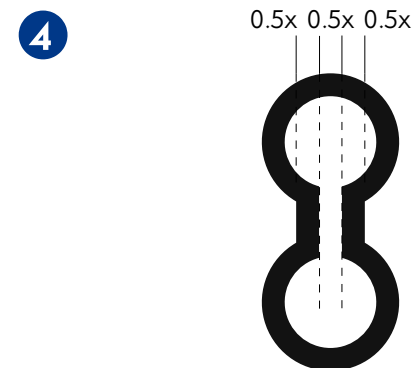
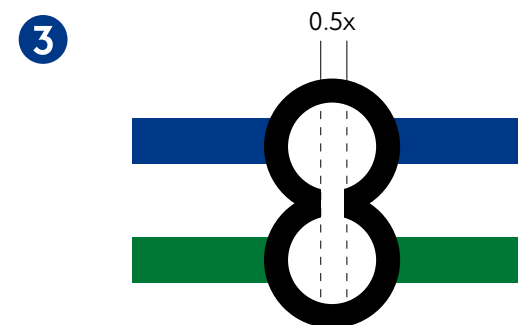
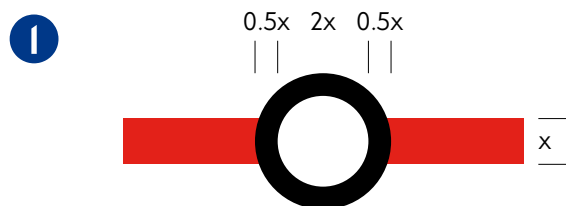
7 Arrow heads

Where lines are terminated beyond the display area of the line diagram, the continuation of the line is denoted by an arrow head.



8 Interchange circles

1. Interchange stations are denoted by circles. These circles are always printed in black with the centre left free of any print
2. Where at a station or stop you have two (or even three) lines sharing the same track and platforms, the interchange circle is centred between the lines
3. Where a customer must change platforms to move on to another line, then more than one interchange circle is to be shown. The circles are drawn so that the outside ring of the circle touches the inside ring of the adjoining circle. A 0.5x white strip is placed across the two circles, thus creating a bridging effect
4. Where necessary, the dumbbell effect may be used to illustrate the interchange between different platforms

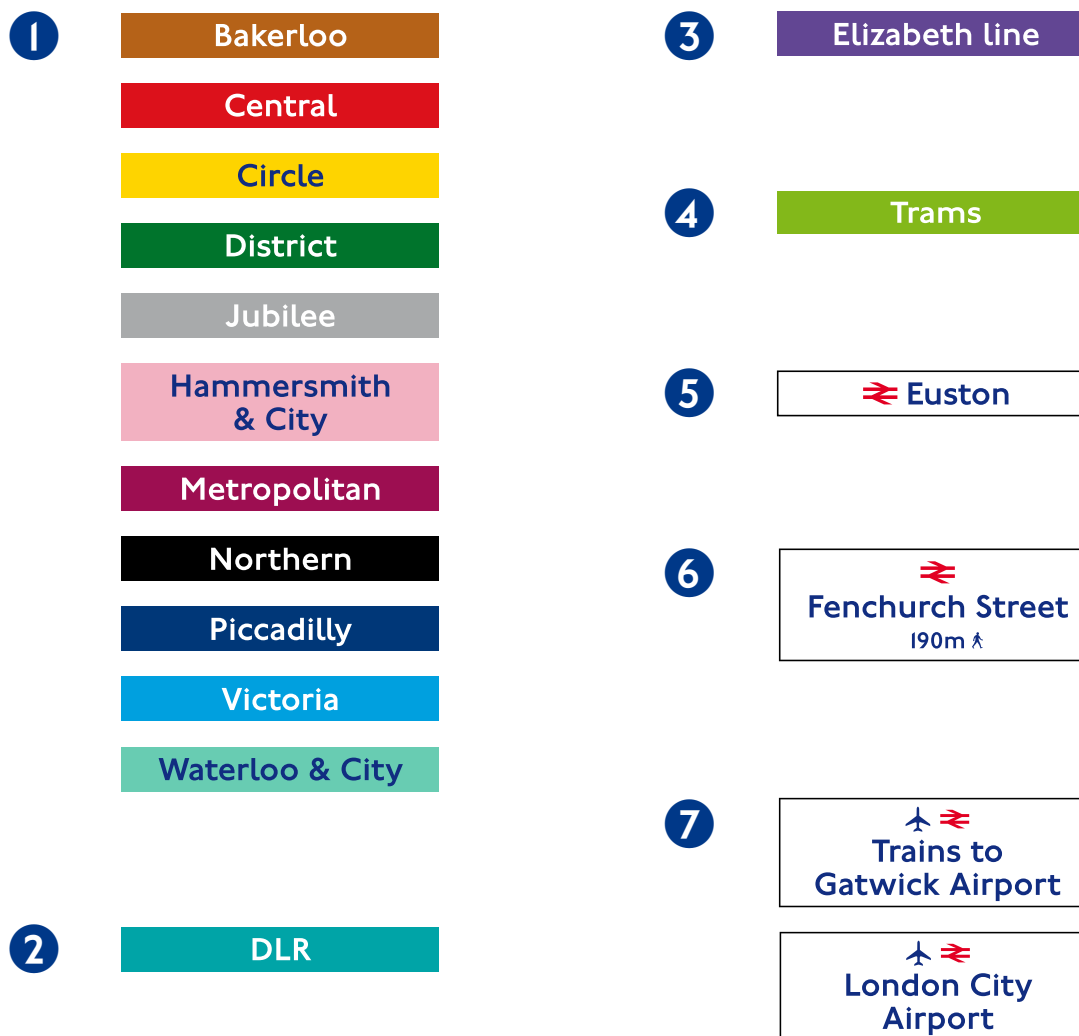


9 Interchange flag boxes and graphic elements

Flag boxes are ordered alphabetically. The first boxes displayed are always Underground lines followed by other TfL modes (again in alphabetical order) and then National Rail stations.

The boxes print in the appropriate line colour, with the type reversed out of white, except for the Circle, Hammersmith & City and Waterloo & City lines where the type is Corporate blue.

1. Interchanges with London Underground lines
2. Interchange with DLR
3. Interchange with Elizabeth line
4. Interchange with London Trams
5. Interchange with National Rail
6. Walking interchange with National Rail
7. Rail interchange with service to airport



9.1 Interchange flag boxes and graphic elements continued

8. Interchange with
London Overground lines

8



9. Walking interchange with
London Overground lines



10. Interchange with London Cable Car



11. Interchange with piers



12. Walking interchanges



10



11



12



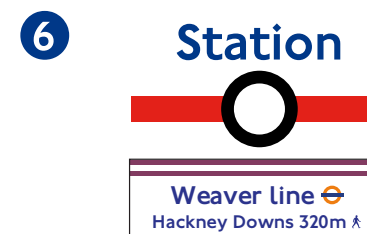
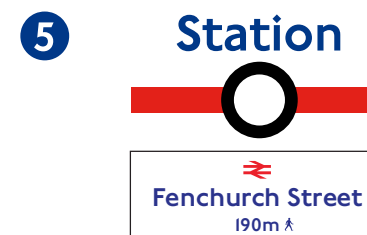
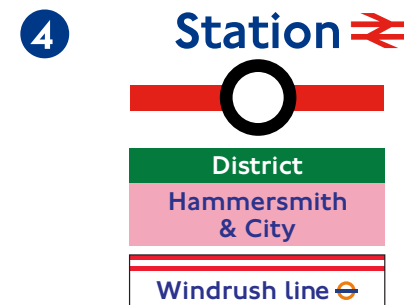
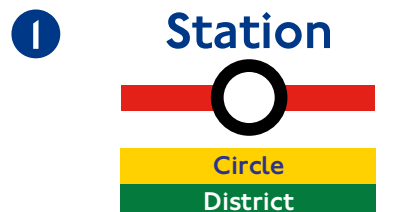
9



9.2 Interchange flag boxes and graphic elements continued

When placed onto a line diagram, the flag boxes and graphic elements are as follows:

1. Interchange with London Underground lines
2. Interchange with National Rail
3. Interchange with a London Overground line (always display National Rail logo at Overground stations)
4. Interchange with London Underground lines and London Overground station
5. Walking interchange with National Rail (interchange ring omitted and rail logo within flag box)
6. Walking interchange with London Overground line (interchange ring omitted and rail logo within flag box)

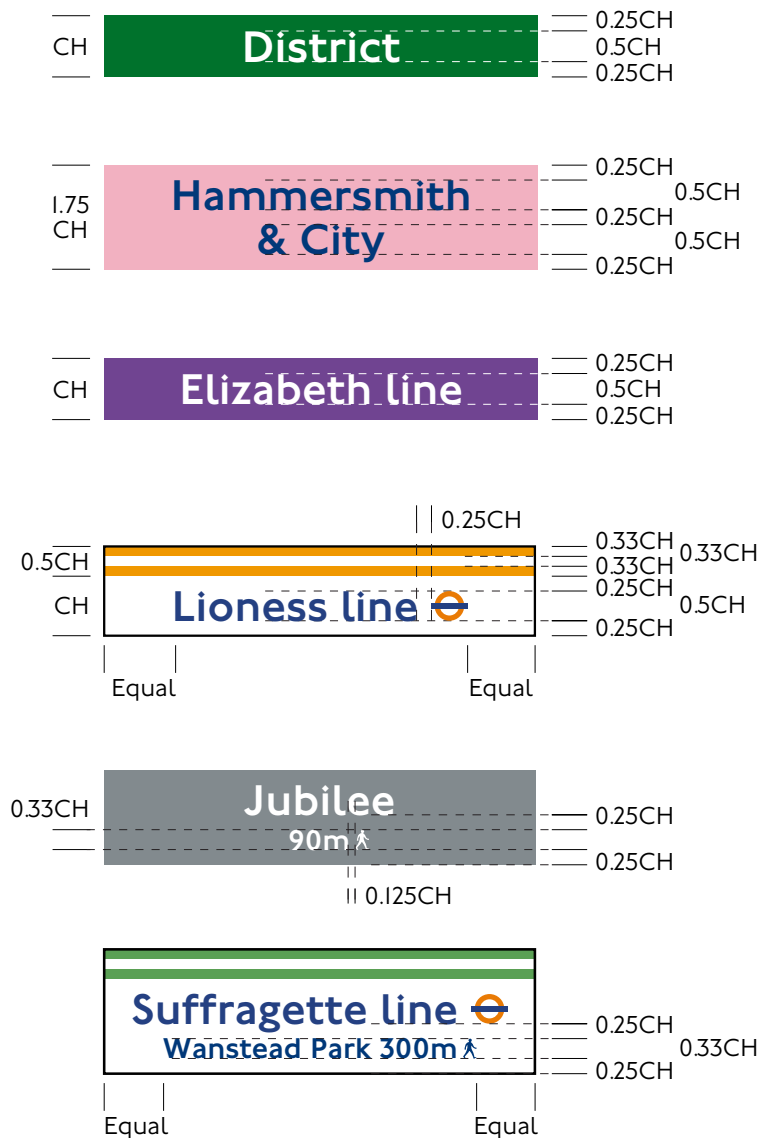


9.3 Interchange flag boxes and graphic elements continued

Flag boxes come in both single-lined or double-lined sizes. The standard height for a single-lined flag box is equal to that of the cap height of the station name. The double-lined versions are constructed as shown.

London Overground line flag boxes come in a single size. The standard height is one and a half times the cap height of the station name.

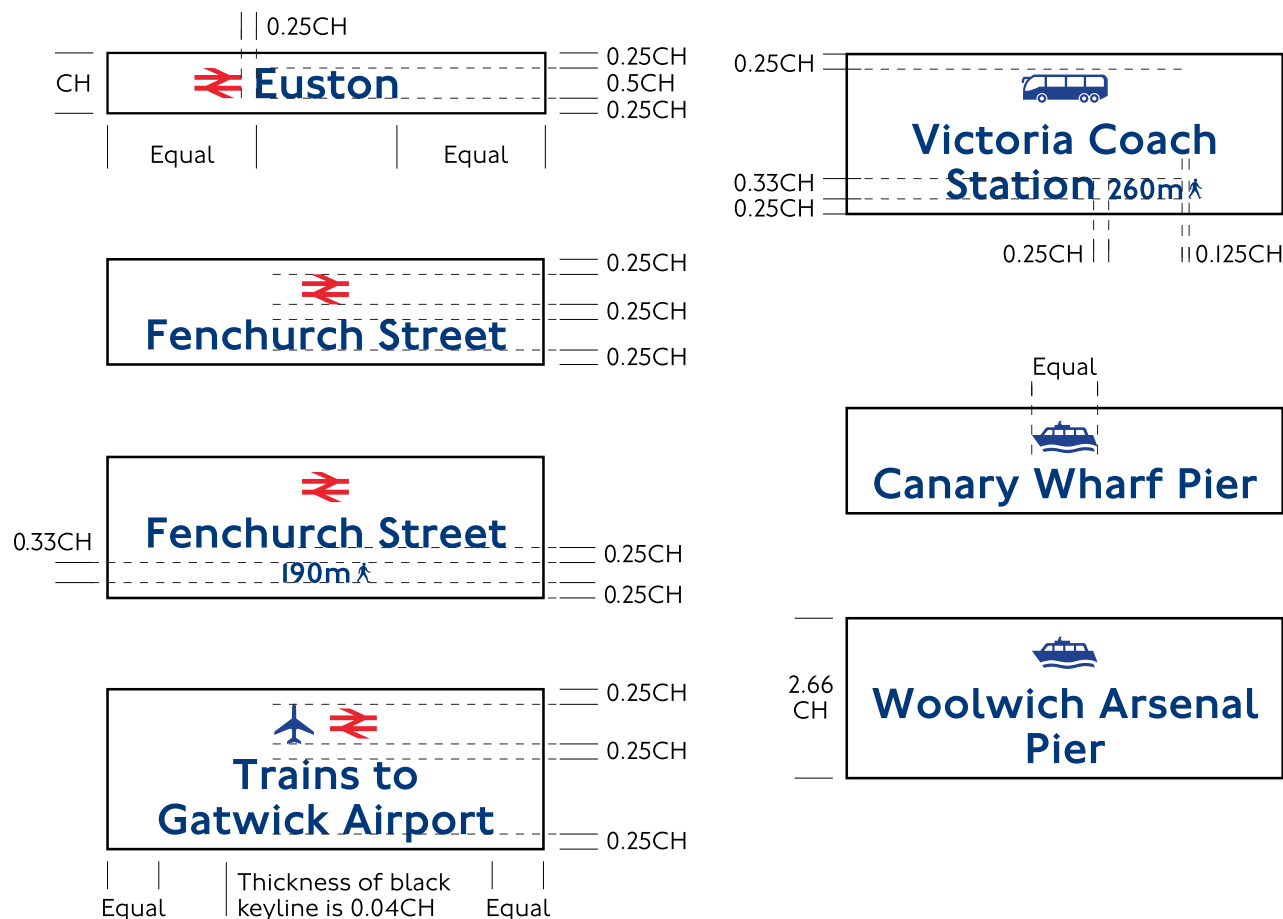
Where there is an interchange within walking distance that is of a different name to the Underground station, the station name and logo appear in an outlined box.



9.4 Interchange flag boxes and graphic elements continued

A connection with National Rail is denoted by the National Rail logo. Where the distance is within walking distance and of a different name to the Underground station, the station name and logo appear in an outlined box.

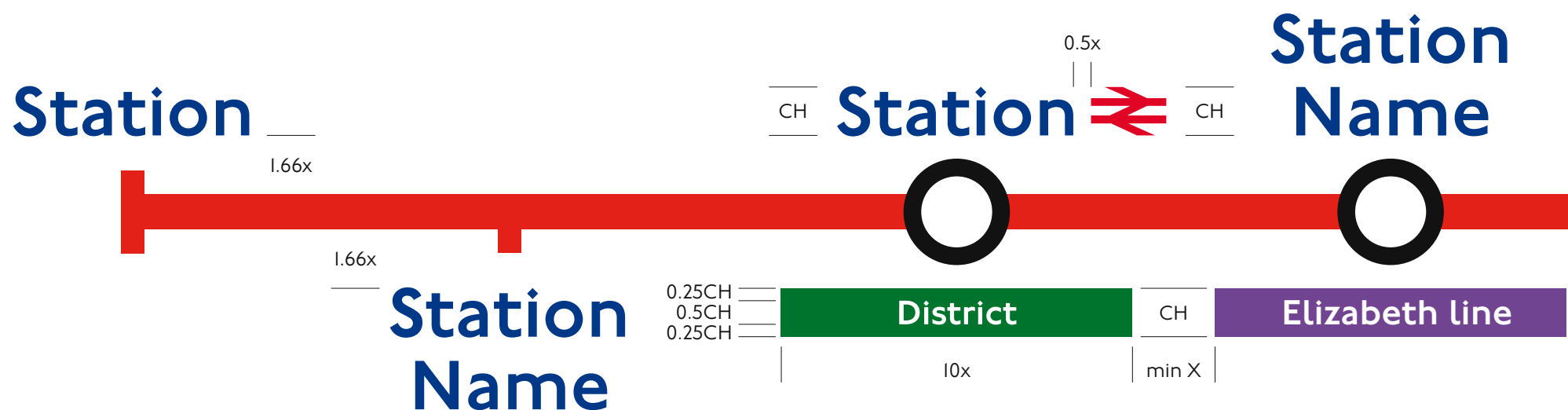
Interchanges with an airport, between piers and at Victoria Coach Station are denoted using the appropriate pictogram appearing in an outlined box.



10 Constructing a line diagram

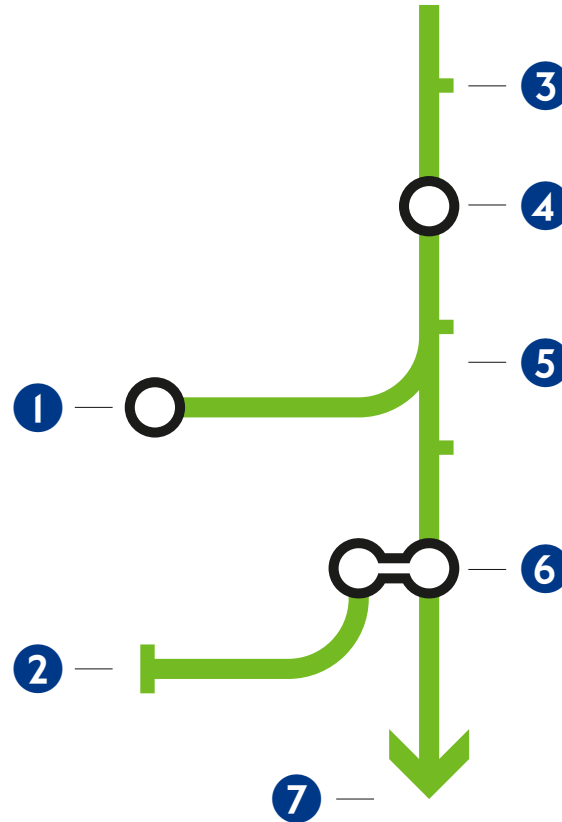
The image below illustrates how the elements are put together to construct a horizontal line diagram.

Flag boxes are always centred beneath the interchange circle and never above. When a flag box is required, the station name always appears above the interchange circle.



10.1 Constructing a line diagram continued

1. Ring to be used where station at the line end is an interchange
2. Double tab is used to indicate the end of the line
3. Single tab is used to represent a station or stop
4. Ring is used to represent an interchange station or stop
5. Direct joined branch indicates a service that can be reached directly as part of a normal service
6. A double ring bridging a main and branch line indicates an indirect branch (this can also be used to indicate a very infrequent service)
7. Arrows are used to indicate that a line continues beyond the stations shown



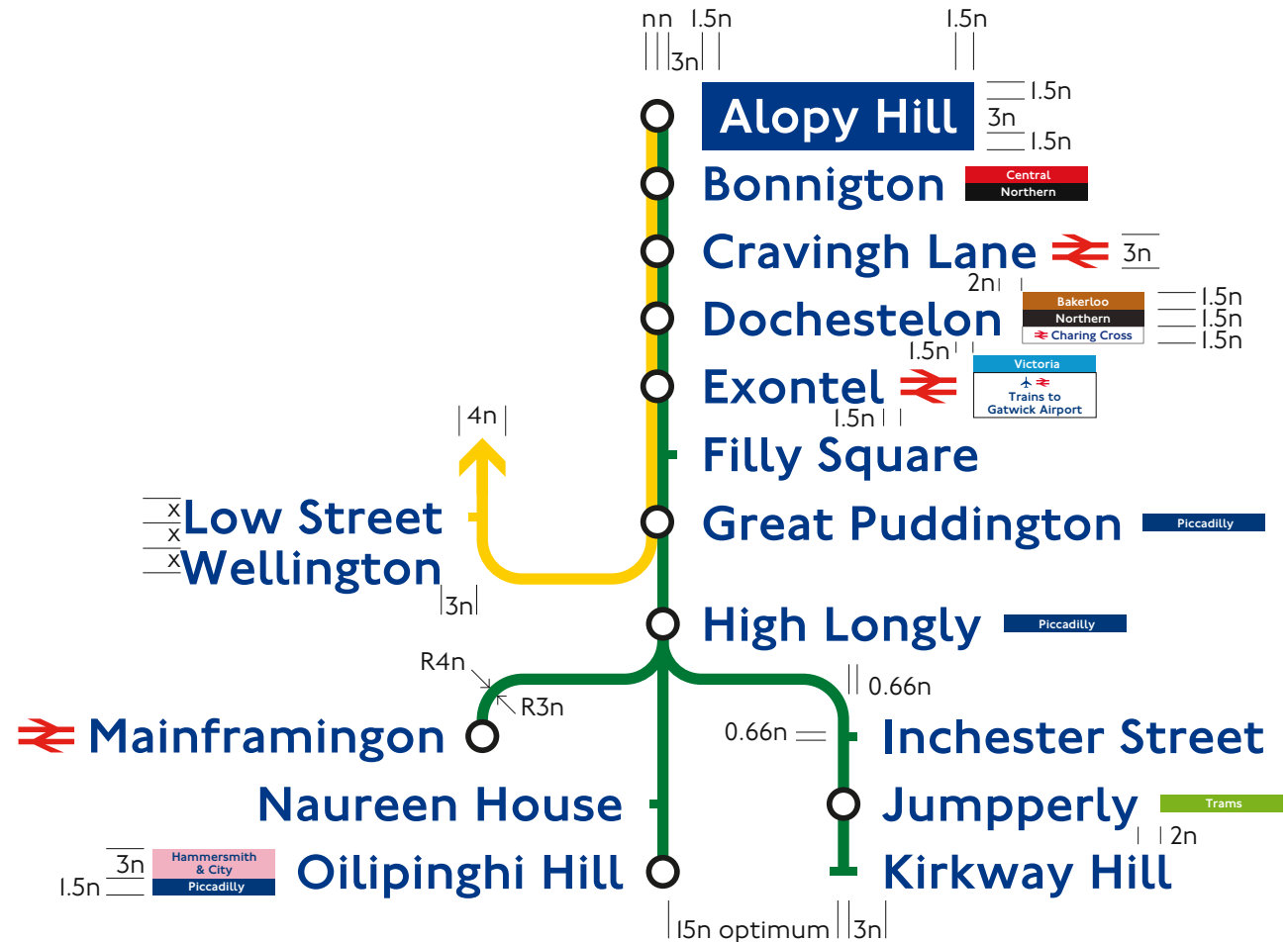
II Vertical platform line diagrams

Vertical platform line diagrams are an integral part of TfL signage systems. They operate slightly differently to the standard line diagram.

The host station name is always to appear first, reversed out of a Corporate blue box. Station names are normally ranged left and to the right of the line. In more complex line diagrams, for example the District and Circle lines, the names may appear on the left where appropriate.

Station name cap heights are 3 x the route line width, while the depth of the flag boxes is equal to 1.5 x the route line width.

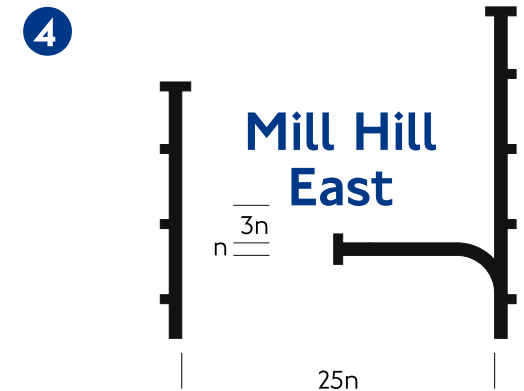
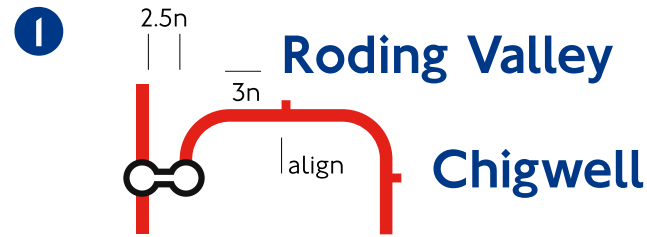
When a vertical platform line diagram is used, the flag box is centred with the cap height of the station name.



11.1 Vertical platform line diagrams continued

Where double circles and branch lines are required for interchanges, the rules shown here, and on the following page, are to be followed.

1. Bridged double circle to denote branch line
2. Branch line with station name shown on the right
3. For triple lines a double ring is used
4. Where two lines run side by side, the space between the two lines is as shown
5. For specific areas such as Heathrow, individual solutions will be established to best suit the situation by the TfL Graphics team



For further information

This standard intends to outline basic principles and therefore cannot cover every application or eventuality. In case of difficulty or doubt as to how to apply these standards, please contact the [TfL Graphics team](#).

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